Riemannian versus KL Discretization of the Gaussian Natural Gradient Flow: Convergence, Cost, and Stepsize Stability

Abstract

The Gaussian natural gradient flow for variational inference, $\dot{m} = Cg$, $\dot{C} = C + CHC$, can be time-discretized in (at least) two ways that share the same explicit mean update but differ in the covariance update: a *Riemannian-distance* (geodesic / exponential-map) step and a *KL/Bregman* (rational inverse-covariance) step. The current analysis of the KL scheme carries a sufficient stepsize condition that is markedly smaller than that of the Riemannian scheme, by a factor $\max\{1, \lambda_{\max}^3/(2\lambda_{\min}^3)\}$ that can be enormous for ill-conditioned covariances. We test whether this smaller KL stepsize bound is a genuine restriction or an artifact of the proof. On three deterministic two-dimensional targets—an exact Gaussian posterior, a non-separable quartic log-concave posterior, and a smooth strongly log-concave posterior—and a scalar diagnostic, we classify every (target, λ, method, Δt) run as SPD-feasible, stable, monotone, or accurate, and compare the empirical maximum stepsizes against the theoretical bounds. The empirical KL stability limit exceeds its theoretical bound by four to six orders of magnitude, far more than the Riemannian scheme's one-to-two orders, indicating the KL stepsize condition is conservative. A scalar diagnostic isolates the cause: the KL covariance update is unconditionally well posed for log-concave targets, and the large-stepsize failures of *both* schemes are driven by the shared explicit mean update, not by the covariance step. Beyond stability, we conduct a comprehensive convergence-speed study across all three targets, three conditionings λ, and every convergent stepsize (Sec. 6): at fine stepsizes the two schemes converge at essentially the same rate in elapsed time $n\Delta t$ and the iteration count to a fixed tolerance scales as $1/\Delta t$, while at the coarsest convergent stepsizes they separate: on the two globally smooth targets the Riemannian (geodesic) update converges faster per step than KL, retaining higher fidelity to the continuous flow, whereas on the non-smooth quartic target KL is the faster scheme.

1 Introduction

We study the Gaussian variational inference problem $\min_{m,C} \mathcal{E}(a)$ with $\mathcal{E}(a) = \mathrm{KL}(\rho_a \,\|\, \rho_{\mathrm{post}})$, $\rho_a = \mathcal{N}(m, C)$, $a = (m, C)$, $C \in \mathrm{SPD}(d)$, and the natural gradient flow that minimizes it in the Fisher–Rao geometry. Writing

$$g_n = \mathbb{E}_{\rho_{a_n}}\big[\nabla_\theta \log \rho_{\mathrm{post}}(\theta)\big], \qquad H_n = \mathbb{E}_{\rho_{a_n}}\big[\nabla_\theta^2 \log \rho_{\mathrm{post}}(\theta)\big], \tag{1}$$

the continuous flow is

$$\dot{m}_t = C_t g_t, \qquad \dot{C}_t = C_t + C_t H_t C_t. \tag{2}$$

Two natural discretizations of (2) share the same explicit mean update and differ only in how the covariance is advanced. The Riemannian-distance discretization takes a true geodesic step of the affine-invariant geometry; the KL/Bregman discretization takes a proximal/mirror step in the dually flat (Bregman) geometry of the exponential family. The two are equally faithful to the continuous flow to first order, but their finite-stepsize behavior, and in particular the sufficient stepsize conditions under which the current analysis proves descent, differ substantially.

The central question is sharp. The Riemannian proof admits stepsizes up to $\Delta t_{\mathrm{Riem,theory}} = 1/(\beta\lambda_{\max})$, whereas the KL proof admits only $\Delta t_{\mathrm{KL,theory}} = 1/\big(\beta\lambda_{\max}\max\{1, \lambda_{\max}^3/(2\lambda_{\min}^3)\}\big)$. The extra factor is 1 only when the relevant covariance eigenvalues are comparable; for the ill-conditioned states that arise from a poorly scaled initialization it is astronomically large, and the KL bound becomes vanishingly small. We ask whether the KL scheme actually *needs* such tiny stepsizes, or whether the restriction is an artifact of the bounding argument—mirroring the dimension-artifact question studied for the local rate in the companion report.

All experiments here are one- or two-dimensional, fully deterministic (closed-form or Gauss–Hermite expectations, no Monte Carlo), and CPU-only, so that the stepsize and stability conclusions are not confounded by sampling error.

2 Algorithms

Both schemes evaluate g_n and H_n once at the current state $a_n = (m_n, C_n)$ and share the explicit mean update

$$m_{n+1} = m_n + \Delta t\, C_n g_n. \tag{3}$$

They differ only in the covariance update.

Riemannian-distance discretization. A geodesic step of the affine-invariant geometry,

$$C_{n+1} = e^{\Delta t}\, C_n^{1/2}\, \exp\big(\Delta t\, C_n^{1/2} H_n C_n^{1/2}\big)\, C_n^{1/2}. \tag{4}$$

This is geometrically faithful—it is exactly the exponential map of the metric along the natural-gradient direction—and it preserves positive definiteness automatically, since it conjugates a matrix exponential (always SPD) by $C_n^{1/2}$. Its cost is a symmetric eigendecomposition (for $C_n^{1/2}$ and the matrix exponential) per step; in the $d \le 2$ regime studied here this is negligible, but in general it is the more expensive update.

KL/Bregman discretization. A proximal step in the dually flat geometry of the Gaussian exponential family,

$$C_{n+1} = (1 + \Delta t)\big(C_n^{-1} - \Delta t\, H_n\big)^{-1}. \tag{5}$$

This is computationally simpler—a single symmetric solve of the precision matrix $P_n = C_n^{-1} - \Delta t\, H_n$—and exposes a structural fact we will use repeatedly: when the target is log-concave, $H_n = -\mathbb{E}[\nabla^2 V] \preceq 0$, so $P_n = C_n^{-1} - \Delta t\, H_n \succ 0$ for *every* $\Delta t > 0$ and the KL covariance update is unconditionally well posed and SPD. Any KL failure must therefore come from elsewhere in the iteration.

Implementation. Both covariance updates symmetrize the result, $C \leftarrow (C + C^\top)/2$, after every step. The Riemannian step uses the eigendecomposition of the symmetric argument for $C^{1/2}$ and exp; the KL step forms C_n^{-1} by a Cholesky solve, builds P_n, and inverts it through its Cholesky factor. We record the extreme eigenvalues of C_n at every step and treat any NaN/Inf, loss of positive definiteness ($\lambda_{\min}(C_n) \le 10^{-12}$), or energy explosion as a failure.

Consistency. Linearizing (4) and (5) in Δt gives, for both, $C_{n+1} = C_n + \Delta t\,(C_n + C_n H_n C_n) + O(\Delta t^2)$, matching $\dot{C}$ in (2); together with (3) both schemes are first-order consistent with the continuous flow. (The test suite verifies this numerically by finite differences.)

3 Targets

We use three two-dimensional targets and one scalar diagnostic. Throughout, $V = -\log \rho_{\text{post}}$ up to a constant, $g = -\mathbb{E}[\nabla V]$, $H = -\mathbb{E}[\nabla^2 V]$, and the energy minimized (the KL up to an a-independent constant) is

$$\mathcal{E}(a) = \mathcal{E}(m, C) = -\tfrac{1}{2} \log \det C + \mathbb{E}_{\mathcal{N}(m,C)}[V(\theta)], \tag{6}$$

whose gap $\mathcal{E}(a_n) - \mathcal{E}(a_\star)$ is the convergence diagnostic plotted throughout. The two-dimensional targets all use $\lambda \in \{0.01, 0.1, 1\}$ and the ill-conditioning is intentional: the proof factor $\lambda_{\max}^3/(2\lambda_{\min}^3)$ is largest exactly when the initial covariance and the target curvature span a wide range of scales.

Gaussian posterior (target A). $\Phi_R(\theta) = \frac{1}{2}\theta^\top Q \theta$ with $Q = \operatorname{diag}(1, \lambda)$, so $\rho_{\text{post}} = \mathcal{N}(0, Q^{-1})$ and

$$g_n = -Q m_n, \qquad H_n = -Q \tag{7}$$

are exact and state-independent. The optimum is exact, $m_\star = 0$, $C_\star = Q^{-1}$, and the energy gap is analytic,

$$\mathcal{E}(a_n) - \mathcal{E}(a_\star) = \tfrac{1}{2}\big[m_n^\top Q m_n + \operatorname{Tr}(QC_n) - \log\det(QC_n) - 2\big]. \tag{8}$$

Initialization $m_0 = (10, 10)^\top$, $C_0 = \operatorname{diag}(\frac{1}{2}, 2)$. With $\nabla^2 V \equiv Q$ the global constants are $\alpha = \min(1, \lambda)$, $\beta = \max(1, \lambda)$. This is the cleanest stepsize test: there is no quadrature or sampling error anywhere.

Non-smooth (quartic) log-concave posterior (target B). $\Phi_R(x, y) = \frac{(\sqrt{\lambda}x - y)^2}{20} + \frac{y^4}{20}$. This is convex but *not* globally smooth: $\nabla^2 V$ grows without bound through the quartic term, so no theorem-based global (α, β) exists and we set `theory_bound_available=false`; only empirical stepsize thresholds are reported. We refer to this target as *non-smooth log-concave* throughout, the defining property being the absence of a global smoothness constant. All required expectations close in the Gaussian moments of Y, $\mathbb{E}[Y^2] = m_y^2 + C_{22}$, $\mathbb{E}[Y^3] = m_y^3 + 3m_y C_{22}$, $\mathbb{E}[Y^4] = m_y^4 + 6m_y^2 C_{22} + 3C_{22}^2$, so g, H and $\mathbb{E}[V]$ are exact. Initialization $m_0 = (10, 10)^\top$, $C_0 = 4I$. The optimum $a_\star$ is found numerically (below).

Smooth strongly log-concave posterior (target C). $\Phi_R^{\text{smooth}}(x, y) = \frac{(\sqrt{\lambda}x - y)^2}{20} + \frac{\delta}{2}(x^2 + y^2) + \gamma \log\cosh(y)$ with $\delta = 0.05$, $\gamma = 1$. This shares the non-separable coupled term of target B but is globally strongly log-concave and smooth, with

$$\alpha = \delta, \qquad \beta = \delta + \tfrac{1+\lambda}{10} + \gamma, \tag{9}$$

from the rank-one coupled quadratic (top eigenvalue $(1+\lambda)/10$), the δI ridge, and $0 \le \gamma \operatorname{sech}^2(y) \le \gamma$. Expectations of $\tanh(Y)$, $\operatorname{sech}^2(Y)$ and $\log\cosh(Y)$ under $Y \sim \mathcal{N}(m_y, C_{22})$ are computed by 80-node Gauss–Hermite quadrature (deterministic, no Monte Carlo). Initialization $m_0 = (10, 10)^\top$, $C_0 = 4I$.

Scalar Gaussian diagnostic (target D). $\rho_{\text{post}} = \mathcal{N}(0, 1)$, so $g = -m$, $H = -1$, and the covariance recursions reduce to the closed forms

$$\text{KL: } C_{n+1} = \frac{(1 + \Delta t) C_n}{1 + \Delta t\, C_n}, \qquad \text{Riem: } C_{n+1} = C_n \exp\big(\Delta t(1 - C_n)\big), \tag{10}$$

with the shared mean step $m_{n+1} = (1 - \Delta t\, C_n) m_n$. The KL form gives $C_{n+1} - 1 = (C_n - 1)/(1 + \Delta t\, C_n)$, a strict contraction toward $C = 1$ for any $\Delta t > 0$. We sweep $C_0 \in \{0.01, 0.1, 10, 100\}$ and $\Delta t \in \{0.5, 1, 2, 5\}$, plus a mean-overshoot probe $m_0 = 1$, $C_0 = 100$.

Reference optimum. For the non-Gaussian targets B and C we minimize (6) over the Cholesky parameterization $C = LL^\top$, $L_{11} = e^{s_1}$, $L_{22} = e^{s_2}$, L_{21} free, with L-BFGS-B and analytic gradients (via the Bonnet/Price identities $\partial_m \mathbb{E}[V] = \mathbb{E}[\nabla V]$, $\partial_C \mathbb{E}[V] = \frac{1}{2}\mathbb{E}[\nabla^2 V]$) from several deterministic restarts. The fitted optima are stationary to $\| C_\star^{-1} + H_\star \|_F < 10^{-9}$; optimizer diagnostics are saved to `target_metadata.json`.

Continuous-time reference. For each target/λ we also integrate (2) from the same initial condition with `solve_ivp` (Radau, rtol $= 10^{-10}$, atol $= 10^{-12}$), flattening the symmetric C state and symmetrizing the vector field. This continuous reference is used only for the discretization *accuracy* metric and the overlaid reference curves; the stability conclusions do not depend on it.

4 Stepsize criteria

We sweep $\Delta t \in \{0.001, 0.002, 0.005, 0.01, 0.02, 0.05, 0.1, 0.2, 0.5, 1, 2, 5, 10\}$ to a fixed terminal time $T = 15$, taking $N = \lceil T/\Delta t \rceil$ steps per run so that $N\Delta t = T$ (up to the final partial step): every run integrates to the same elapsed time T, and the elapsed time after step n is $n\Delta t$. The coarsest stepsizes ($\Delta t = 5, 10$) are deliberate very-coarse stability probes. Each run is classified from its *full* trajectory (the long-format CSV may be decimated for size, but classification always uses every step).

Definition 1 (Feasibility classes)**.** With $\mathcal{E}_n = \mathcal{E}(a_n)$ and $\mathcal{E}_\star = \mathcal{E}(a_\star)$, a run is

- **SPD-feasible** if it has no NaN/Inf and $\lambda_{\min}(C_n) > 10^{-12}$ for all n;
- **stable** if SPD-feasible, $\mathcal{E}_N < \mathcal{E}_0$, and there is no explosion, $\mathcal{E}_n - \mathcal{E}_\star \le 10^3(\mathcal{E}_0 - \mathcal{E}_\star)$ for all n;
- **monotone** if SPD-feasible and $\mathcal{E}_{n+1} \le \mathcal{E}_n + 10^{-10}$ for all n;
- **accurate** if SPD-feasible and the terminal discretization error against the continuous reference satisfies $\|m_N - m(T)\|_2 + \|C_N - C(T)\|_F / \|C(T)\|_F \le 10^{-2}$.

For each (target, λ, method) we report the largest feasible stepsize in each class, $\Delta t_{\max}^{\mathrm{SPD}}$, $\Delta t_{\max}^{\mathrm{stable}}$, $\Delta t_{\max}^{\mathrm{monotone}}$, $\Delta t_{\max}^{\mathrm{accurate}}$ (a plain maximum over passing stepsizes, so a non-monotone feasibility boundary is captured faithfully).

Definition 2 (Theoretical stepsizes)**.** For the globally smooth targets (A, C), set $\lambda_{\min} = \min\{\lambda_{0,\min}, 1/\beta\}$ and $\lambda_{\max} = \max\{\lambda_{0,\max}, 1/\alpha\}$ from the extreme eigenvalues of C_0. The proof Lipschitz constants and sufficient stepsizes are

$$\Delta t_{\mathrm{Riem,theory}} = \frac{1}{\beta\lambda_{\max}}, \qquad \Delta t_{\mathrm{KL,theory}} = \frac{1}{\beta\lambda_{\max}\,\max\{1,\ \lambda_{\max}^3/(2\lambda_{\min}^3)\}}. \tag{11}$$

For the non-smooth target B (no global smoothness constant) no such bound is reported.

The diagnostic quantities are the ratios $\Delta t_{\max}^{\mathrm{stable}}/\Delta t_{\mathrm{theory}}$ and $\Delta t_{\max}^{\mathrm{monotone}}/\Delta t_{\mathrm{theory}}$: a ratio close to 1 means the proof bound is tight; a ratio of many orders of magnitude means the bound is conservative. Because the KL theoretical stepsize carries the extra $\lambda_{\max}^3/(2\lambda_{\min}^3)$ factor, its ratio is the focus of the proof-artifact discussion.

5 Results

Metadata. Table 1 lists the reference optimum $\mathcal{E}_\star$ and the global constants (α, β) for each target/λ. The Gaussian and smooth targets are globally smooth (theory bounds available); the non-smooth target is not.

Table 1: Per-target reference optimum $\mathcal{E}_\star$ and global strong-convexity / smoothness constants (α, β). The non-smooth target is convex but not globally smooth, so no theorem-based (α, β) is reported.

target	λ	α	β	$\mathcal{E}_\star$	global smooth
Gaussian posterior	0.01	0.010	1.000	-1.3026	yes
Gaussian posterior	0.1	0.100	1.000	-0.1513	yes
Gaussian posterior	1	1.000	1.000	1.0000	yes
smooth log-concave	0.01	0.050	1.151	-0.5637	yes
smooth log-concave	0.1	0.050	1.160	-0.4933	yes
smooth log-concave	1	0.050	1.250	-0.0749	yes
non-smooth log-concave	0.01	–	–	-2.8316	no
non-smooth log-concave	0.1	–	–	-1.6803	no
non-smooth log-concave	1	–	–	-0.5290	no

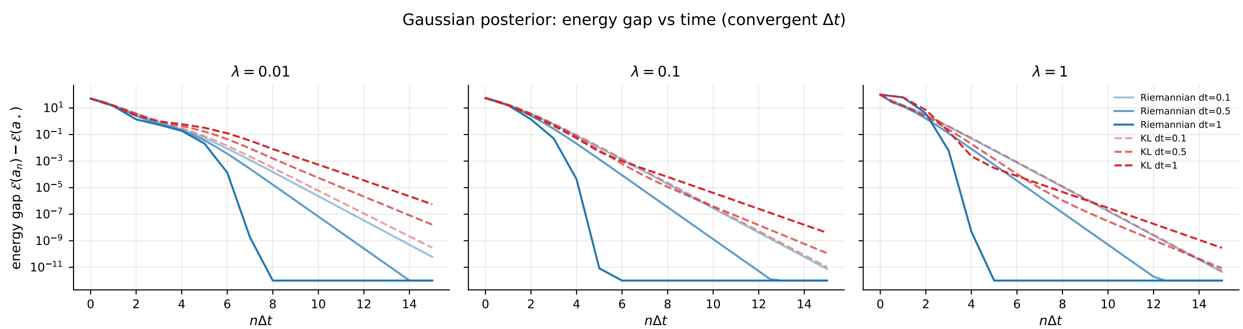

Figure 1: Gaussian posterior: energy gap vs elapsed time $n\Delta t$, $\lambda \in \{0.01, 0.1, 1\}$, Riemannian (blue) vs KL (red) at $\Delta t \in \{0.1, 0.5, 1\}$ (darker = larger Δt). The analytic gap (8) reaches machine precision.

Convergence. Figures 1–2 show the energy gap $\mathcal{E}(a_n) - \mathcal{E}(a_\star)$ against elapsed time $n\Delta t$ for representative stepsizes $\Delta t \in \{0.1, 0.5, 1\}$. On the two globally smooth targets both schemes converge cleanly at these stepsizes; the curves of the two methods are visually close at matched Δt, since they agree to first order and the shared mean update dominates the transient. The Gaussian target reaches the analytic optimum to machine precision; the smooth target converges to the numerically optimized $a_\star$. On the non-smooth target, both schemes require much smaller stepsizes (Fig. 3), consistent with its unbounded curvature: the representative stepsizes shown there are the largest that converge for that target ($\Delta t \in \{0.005, 0.01, 0.02\}$), an order of magnitude below those of the smooth targets.

Stepsize stability. Figures 4–6 are the stability heatmaps: for each $(\lambda, \Delta t)$ a cell is green (stable and monotone), amber (stable but non-monotone), or red (unstable). For the two smooth targets the dotted line marks the theoretical stepsize Δt_{theory} for that method. Two features are immediate. First, for the KL scheme the green/amber region extends *orders of magnitude* to the right of the dotted theoretical line—the empirical stability boundary sits near $\Delta t \approx 1$–2 while the proof admits only $\Delta t \approx 10^{-5}$–10^{-6}. Second, the Riemannian theoretical line sits much closer to its empirical boundary. The same picture holds across all three λ and both smooth targets.

Theory vs empirical (the proof-artifact figure). Figure 7 and Table 2 compare, for the two smooth targets, the theoretical stepsize against the empirical stable, monotone, and accurate

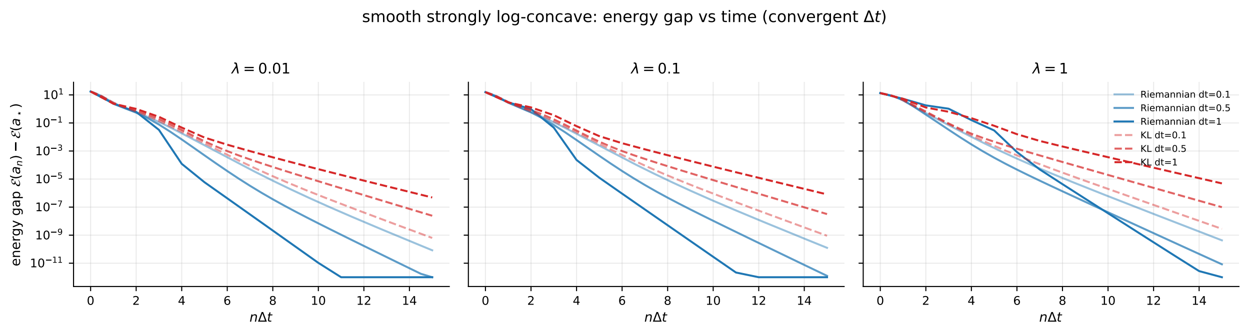

Figure 2: Smooth strongly log-concave posterior: energy gap vs elapsed time $n\Delta t$. Both schemes converge at the representative stepsizes; the gap is measured against the numerically optimized $a_\star$.

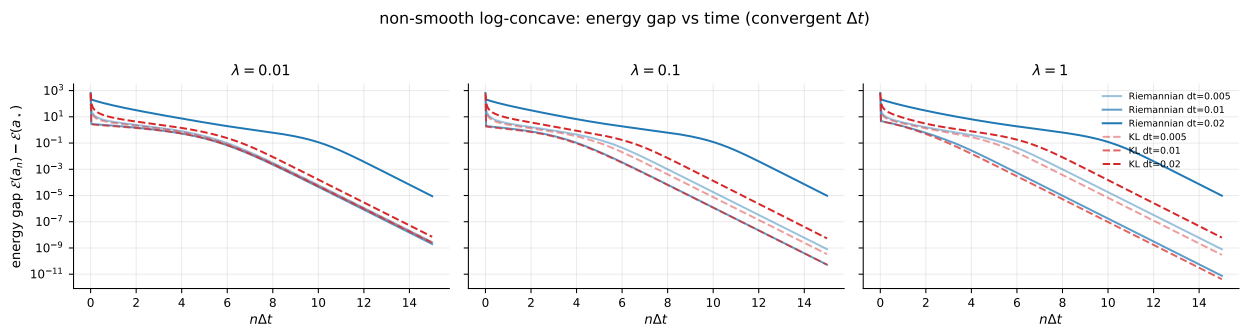

Figure 3: Non-smooth (quartic) log-concave posterior: energy gap vs elapsed time $n\Delta t$ at its convergent representative stepsizes $\Delta t \in \{0.005, 0.01, 0.02\}$. Both schemes converge cleanly here; the stepsizes are an order of magnitude smaller than for the smooth targets, reflecting the unbounded curvature of the quartic tail. (Larger stepsizes diverge for both methods and are excluded; see the stability heatmap, Fig. 6.)

maxima. We emphasize the *monotone* and *accurate* classes over bare stability: monotonicity certifies genuine descent at every step, and accuracy certifies that the discrete trajectory still tracks the continuous flow to 10^{-2}, so both are stronger evidence than non-divergence alone. The empirical KL monotone stepsize exceeds its theoretical bound by ratios from 6.4×10^1 (Gaussian, $\lambda = 1$) up to 8×10^8 (Gaussian, $\lambda = 0.01$), with the smooth target consistently around 1.4–2.0×10^5; the *accurate* maximum is one to two grid steps smaller but still exceeds the KL bound by four to eight orders of magnitude, so the proof-artifact gap is not an artifact of the permissive stability criterion—it persists under the strict accuracy criterion. The Riemannian ratios, by contrast, range only from 2 to 200 (monotone) and are of the same order under accuracy. The KL bound is thus conservative by four to six extra orders of magnitude relative to the Riemannian bound—precisely the size of the $\lambda_{\max}^3/(2\lambda_{\min}^3)$ factor that separates the two theoretical stepsizes.

Scalar diagnostic. Figure 8 traces the scalar covariance recursions (10) from $C_0 \in \{0.01, 0.1, 10, 100\}$ at $\Delta t \in \{0.5, 1, 2, 5\}$. The KL covariance update contracts to $C = 1$ for every C_0 and every stepsize shown—as the closed form $C_{n+1} - 1 = (C_n - 1)/(1 + \Delta t\, C_n)$ guarantees—never leaving the SPD cone. The Riemannian scalar update $C_{n+1} = C_n \exp(\Delta t(1 - C_n))$ can overshoot for large $C_0 \Delta t$. The mean-overshoot probe ($m_0 = 1$, $C_0 = 100$) confirms the companion fact: the shared explicit mean step $m_{n+1} = (1 - \Delta t\, C_n) m_n$ amplifies the mean for both schemes once $\Delta t\, C_n > 2$ (e.g. $|m|$ grows past 10^4 at $\Delta t = 5$ even though the KL covariance remains perfectly stable). This locates the large-stepsize instability in the mean update shared by both schemes, not in the KL covariance step.

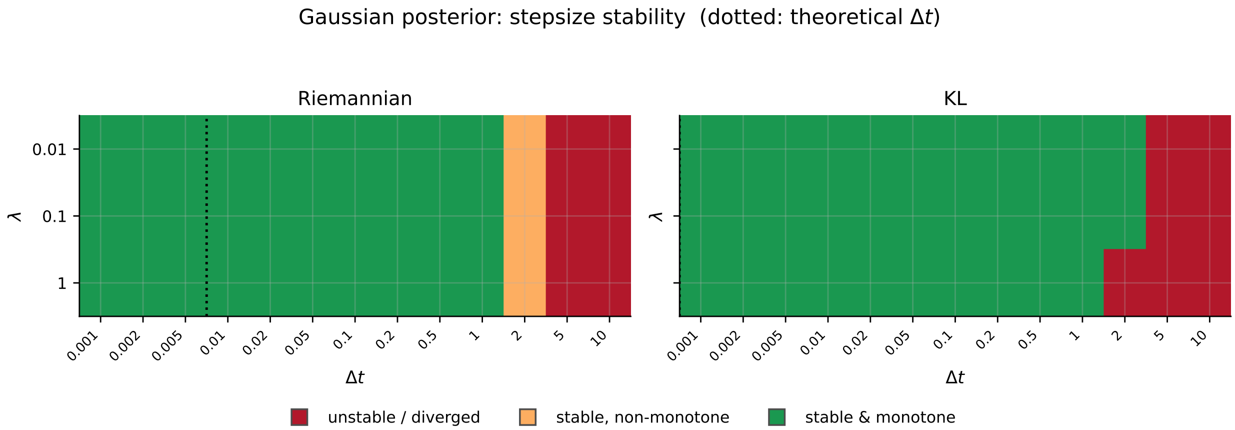

Figure 4: Gaussian posterior stability classification. Green: stable & monotone; amber: stable, non-monotone; red: unstable/diverged. Dotted line: theoretical Δt for the method. The KL stable region extends far past its (tiny) theoretical stepsize.

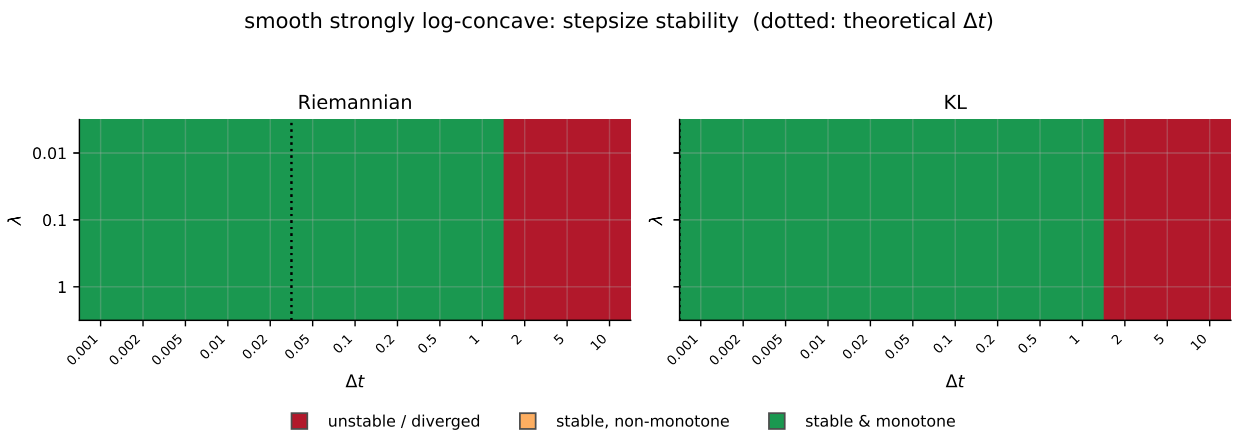

Figure 5: Smooth log-concave stability classification (same legend). The KL theoretical line ($\sim 10^{-6}$) is far to the left of the empirical boundary (~ 1).

Cost. Since the dimension is $d \leq 2$, wall-clock differences between the schemes are small and noisy and we do not draw strong cost conclusions from them. Figure 9 reports iterations-to-tolerance (gap $\leq 10^{-4}$) for stable runs as the meaningful efficiency axis: at matched Δt the two schemes need a comparable iteration count, and the practical advantage of a scheme is governed by how large a stable Δt it admits rather than by per-step cost. The KL covariance step (one symmetric solve) is the cheaper update per step; the Riemannian step (eigendecomposition for $C^{1/2}$ and exp) is more expensive in higher dimension, though that difference is immaterial here.

6 Convergence speed: Riemannian versus KL at matched stepsize

The preceding sections concern *stability*: how large a stepsize each scheme tolerates and whether the KL proof bound is conservative. We now address the complementary and practically decisive question of *convergence speed*: given a stepsize that converges for both schemes, which scheme reaches the optimum faster, and how does that depend on the target, the conditioning λ, and

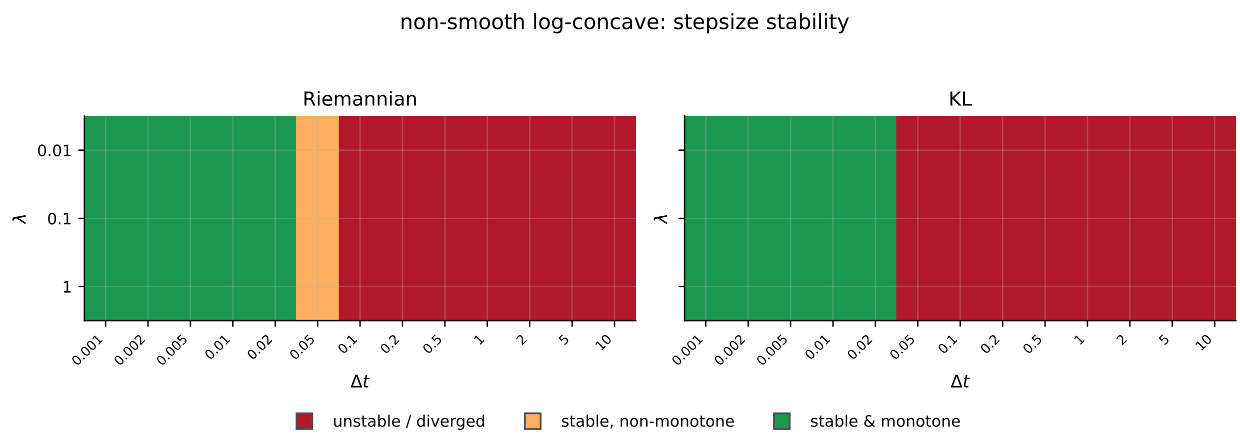

Figure 6: Non-smooth log-concave stability classification (no theory line, since the target is not globally smooth). Both schemes are stable only for small Δt.

the stepsize? This is the question a practitioner choosing between the two updates actually faces. Throughout we restrict to the convergent (stable) stepsizes for each $(\text{target}, \lambda, \text{method})$; divergent stepsizes are excluded by construction, since a diverging run has no meaningful convergence speed.

Summary: Riemannian usually wins at a matched stepsize. Table 3 summarizes the head-to-head comparison. For each target and stepsize it reports the terminal energy gap of each scheme (geometric mean over the three λ), the faster scheme, and the factor separating them. The pattern is clear and is the main practical message of this section. On the two *globally smooth* targets the Riemannian scheme wins at every stepsize, and its advantage *grows with the stepsize*: from a modest 1.3–7$\times$ at the finest stepsizes to 10^4–$10^8\times$ smaller terminal gap at the coarsest convergent stepsizes. Only on the non-smooth quartic target—whose unbounded curvature drives the Riemannian exponential map near its stability edge at the coarsest stable stepsize—does KL win, and there only by a modest factor. So in the regime where one routinely operates (a globally smooth target, a stepsize chosen as large as stability allows), the geodesic Riemannian update is the faster of the two, often by many orders of magnitude.

Energy gap at matched stepsizes. Figures 10–12 show the comparison panel by panel: one figure per target, with rows indexing the conditioning λ and columns indexing a representative convergent stepsize, so that each panel contains exactly two curves—Riemannian (solid) and KL (dashed)—at one $(\lambda, \Delta t)$ cell. The stepsizes shown are the curated convergent set for each target (stable for both schemes at every λ); the non-smooth target uses stepsizes an order of magnitude smaller, reflecting its unbounded curvature. On the two *globally smooth* targets (Figs. 10–11) the comparison splits cleanly by stepsize regime. At the *fine* stepsizes (left columns) the Riemannian and KL curves are nearly indistinguishable: the two schemes agree to first order, so when Δt is small they descend along essentially the same gap-versus-$n\Delta t$ curve, and that curve is the same across columns—the discrete trajectories track the common continuous flow. At the *coarse* stepsizes (right columns), however, the schemes separate visibly: the Riemannian update converges markedly faster per unit elapsed time $n\Delta t$, reaching machine precision while KL plateaus several orders of magnitude higher (at the Gaussian target, $\lambda = 0.01$, $\Delta t = 1$, the terminal gaps are $\sim 10^{-14}$ for Riemannian against $\sim 10^{-7}$ for KL). The geodesic covariance update therefore retains higher-order fidelity to the flow as the stepsize coarsens, whereas the rational KL update accumulates a larger

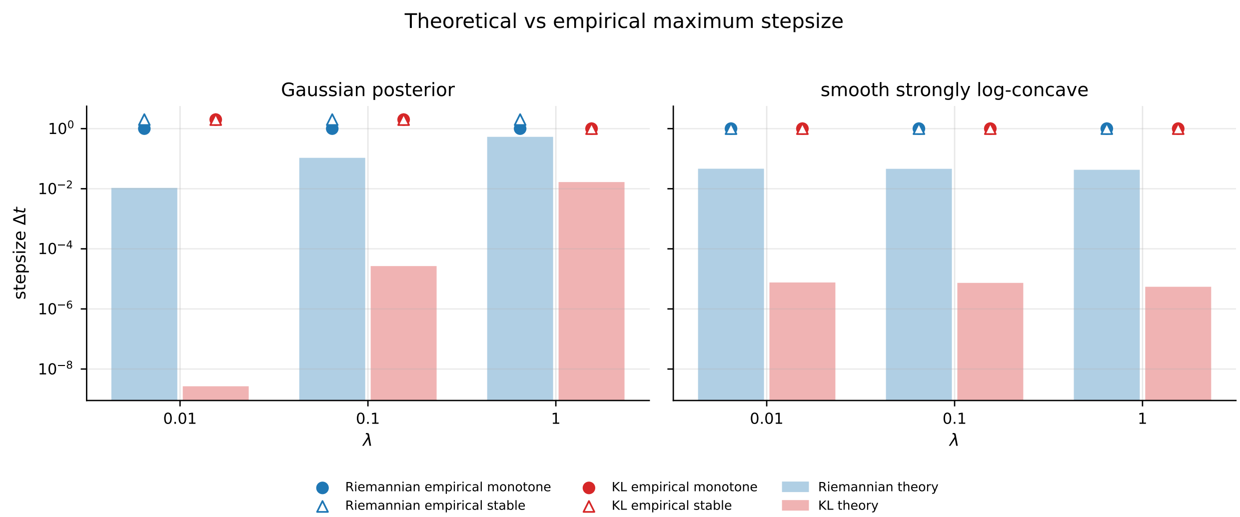

Figure 7: Theoretical (shaded bars) vs empirical maximum stepsize (markers: $\circ$ monotone, $\triangle$ stable) for the Gaussian and smooth targets, Riemannian (blue) vs KL (red), log scale. The gap between the KL theoretical bar and its empirical markers is enormous; the Riemannian gap is small. This is the central proof-artifact figure.

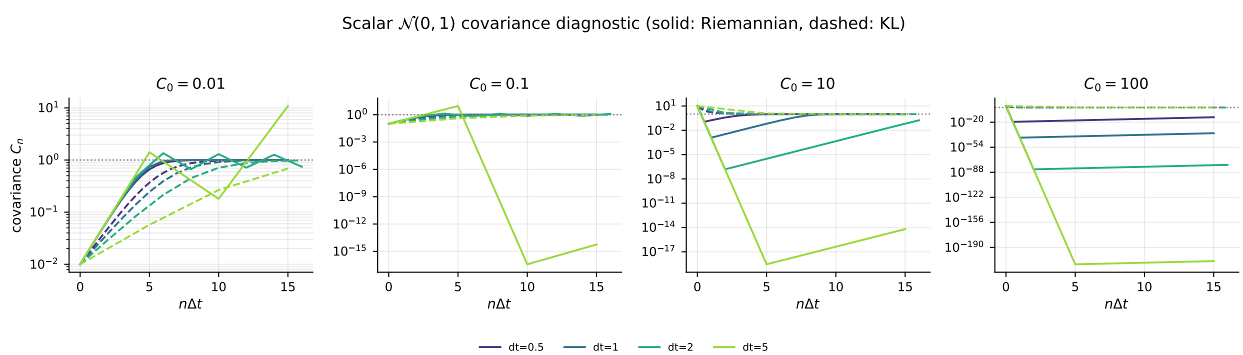

Figure 8: Scalar $\mathcal{N}(0, 1)$ covariance diagnostic. Solid: Riemannian; dashed: KL; color shade encodes Δt; dotted grey: target $C = 1$. The KL covariance contracts to 1 for all $(C_0, \Delta t)$ shown.

discretization error—a genuine per-step convergence-rate advantage for the Riemannian scheme at large stepsizes, distinct from the stability question of the preceding sections. The non-smooth quartic target (Fig. 12) is the exception: there the two schemes agree at the finer stepsizes but *KL* is the faster scheme, decisively so at the coarsest convergent $\Delta t = 0.02$, where the Riemannian exponential map sits near its stability edge while the rational KL update has already converged. In all cases both schemes remain monotone and convergent throughout; the difference is the rate, not whether they converge.

The cost of a stepsize. Figure 13 quantifies the efficiency consequence. For the stable runs we measure the iterations and the elapsed time $n\Delta t$ required to reach a fixed energy-gap tolerance (10^{-6}), averaged over λ, as a function of Δt. The iteration count (top row) falls as $1/\Delta t$ on the log–log axes—a larger convergent stepsize reaches the same tolerance in proportionally fewer steps—while the elapsed time $n\Delta t$ to that tolerance (bottom row) is nearly flat in Δt until the largest convergent stepsizes, where the discrete trajectory begins to lag the continuous flow. Consistent

Table 2: Empirical vs theoretical maximum stepsize by target/λ/method, reporting the largest stable, monotone, and accurate stepsizes and the monotone/theory and accurate/theory ratios. Both ratios are 10^4–10^8 for KL but only 2–10^2 for the Riemannian scheme; the gap survives the strict accuracy criterion, not just bare stability.

target	λ	method	Δt_{theory}	$\Delta t^{\text{stab}}_{\text{max}}$	$\Delta t^{\text{mono}}_{\text{max}}$	$\Delta t^{\text{acc}}_{\text{max}}$	$\frac{\text{mono}}{\text{theory}}$ / $\frac{\text{acc}}{\text{theory}}$
Gaussian	0.01	kl	2.5×10^{-9}	2.000	2.000	1.000	8.0×10^8 / 4.0×10^8
Gaussian	0.01	riemannian	0.01	2.000	1.000	1.000	100 / 100
Gaussian	0.1	kl	2.5×10^{-5}	2.000	2.000	1.000	8.0×10^4 / 4.0×10^4
Gaussian	0.1	riemannian	0.1	2.000	1.000	1.000	10 / 10
Gaussian	1	kl	0.0156	1.000	1.000	1.000	64 / 64
Gaussian	1	riemannian	0.5	2.000	1.000	1.000	2 / 2
smooth	0.01	kl	7.1×10^{-6}	1.000	1.000	1.000	1.4×10^5 / 1.4×10^5
smooth	0.01	riemannian	0.0434	1.000	1.000	1.000	23 / 23
smooth	0.1	kl	6.9×10^{-6}	1.000	1.000	1.000	1.4×10^5 / 1.4×10^5
smooth	0.1	riemannian	0.0431	1.000	1.000	1.000	23.2 / 23.2
smooth	1	kl	5.1×10^{-6}	1.000	1.000	1.000	2.0×10^5 / 2.0×10^5
smooth	1	riemannian	0.04	1.000	1.000	1.000	25 / 25
non-smooth	0.01	kl	–	0.020	0.020	0.020	–
non-smooth	0.01	riemannian	–	0.050	0.020	0.010	–
non-smooth	0.1	kl	–	0.020	0.020	0.020	–
non-smooth	0.1	riemannian	–	0.050	0.020	0.010	–
non-smooth	1	kl	–	0.020	0.020	0.020	–
non-smooth	1	riemannian	–	0.050	0.020	0.020	–

with the matched-stepsize panels, the two schemes coincide at the fine and intermediate stepsizes and separate only at the coarsest convergent stepsizes, where the faster scheme reaches the tolerance in fewer iterations (and the slower one may not reach 10^{-6} at all within the horizon, terminating at a larger plateau). Which scheme that is depends on the target: on the two smooth targets it is Riemannian, while on the non-smooth target it is KL—at $\Delta t = 0.02$ the Riemannian run never reaches 10^{-6} whereas KL does. The practical reading is twofold: the dominant efficiency lever remains how large a stable stepsize a scheme admits (the conclusion of the stability sections), but among the stepsizes both schemes can take, the geometrically faithful Riemannian update converges faster per step in the coarse-stepsize regime on smooth targets, while the two are interchangeable when Δt is small.

7 Conclusion

The evidence supports a balanced reading, not a simple ranking.

- **The Riemannian discretization is the geometrically faithful one.** Its update (4) is exactly a geodesic step of the affine-invariant geometry, with the cleanest interpretation as geodesic gradient descent, and its theoretical stepsize bound is comparatively tight (empirical/theory ratio 2–200). Its cost is the matrix exponential, immaterial here but the more expensive update in general.
- **The KL discretization is computationally simpler and empirically stable far beyond its proof bound.** The update (5) is a single symmetric solve, and on the globally smooth

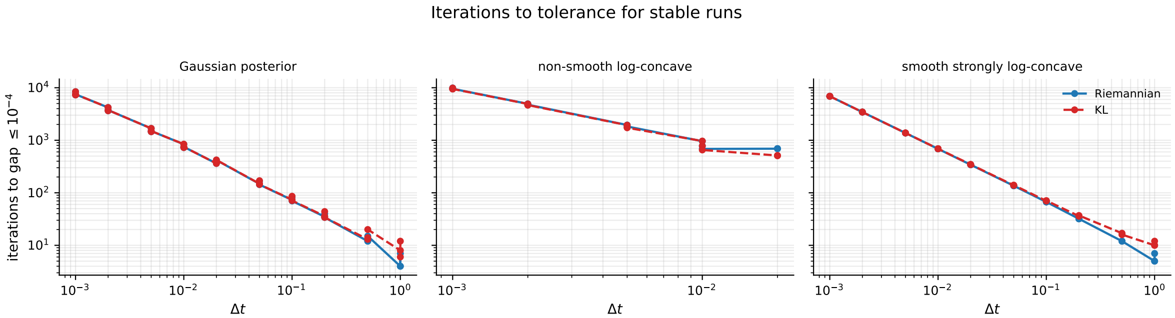

Figure 9: Iterations to reach gap $\leq 10^{-4}$ for stable runs, vs Δt (log–log). At matched Δt the two schemes are comparable; the meaningful difference is the range of stable Δt, not per-step cost.

Table 3: Head-to-head convergence speed at matched stepsize: terminal energy gap $\mathcal{E}(a_N) - \mathcal{E}(a_\star)$ (geometric mean over $\lambda \in \{0.01, 0.1, 1\}$) for each scheme, the faster scheme, and the factor by which its terminal gap is smaller. Riemannian wins on both smooth targets, with the margin growing as Δt coarsens; KL wins only on the non-smooth quartic, and only modestly.

target	Δt	gap_{Riem}	gap_{KL}	winner	factor
Gaussian	0.05	2.5×10^{-11}	3.4×10^{-11}	Riem.	1.33
	0.1	1.3×10^{-11}	2.5×10^{-11}	Riem.	2
	0.5	3.4×10^{-15}	2.5×10^{-10}	Riem.	7.3×10^{4}
	1	1.3×10^{-16}	8.6×10^{-9}	Riem.	6.6×10^{7}
smooth	0.05	2.5×10^{-10}	6.8×10^{-10}	Riem.	2.67
	0.1	1.6×10^{-10}	1.2×10^{-9}	Riem.	7.12
	0.5	2.0×10^{-12}	4.2×10^{-8}	Riem.	2.2×10^{4}
	1	3.0×10^{-15}	1.2×10^{-6}	Riem.	4.1×10^{8}
non-smooth	0.002	2.1×10^{-9}	1.7×10^{-9}	KL	1.29
	0.005	1.2×10^{-9}	6.4×10^{-10}	KL	1.91
	0.01	9.2×10^{-11}	8.1×10^{-11}	KL	1.14
	0.02	9.4×10^{-6}	6.2×10^{-9}	KL	1.5×10^{3}

targets it is stable and monotone for stepsizes 10^4–10^8 times larger than the current sufficient condition permits (Table 2, Fig. 7).

- **The KL stepsize restriction is conservative.** The large and systematic gap between the empirical and theoretical KL limits—of the same order as the $\lambda_{\max}^3/(2\lambda_{\min}^3)$ factor that distinguishes the two proofs—indicates that this factor is an artifact of the bounding argument rather than a genuine restriction on the scheme. It persists under the strict accuracy criterion, not just bare stability (Table 2).

- **The KL covariance update is not the instability source.** For a log-concave target the KL precision $C^{-1} - \Delta t\, H$ is SPD for every $\Delta t > 0$, so the covariance step never leaves the SPD cone; the scalar diagnostic (Fig. 8) confirms unconditional contraction of the covariance to its fixed point.

- **Large-stepsize failures come from the shared mean update.** Both schemes diverge at large Δt through the explicit mean step $m_{n+1} = (1 - \Delta t\, C_n) m_n$, which overshoots once

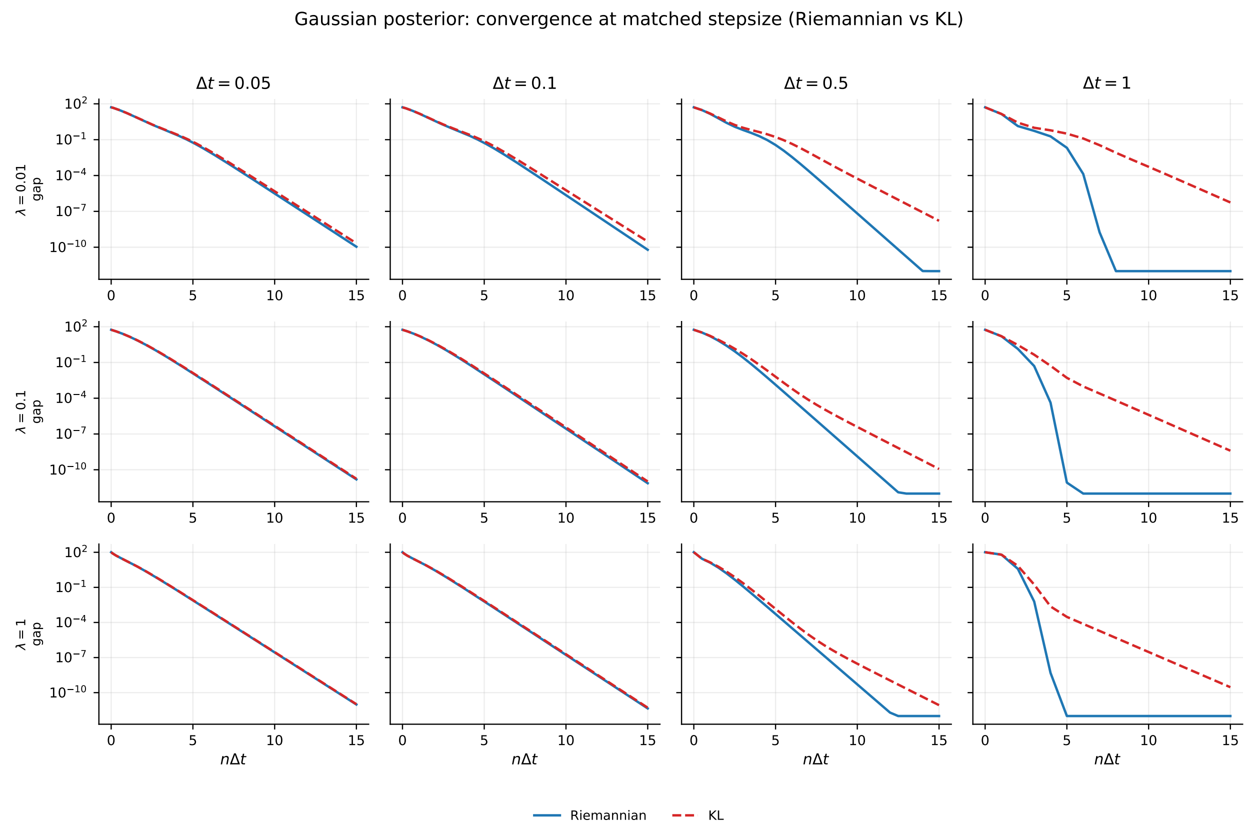

Figure 10: Gaussian posterior: energy gap vs elapsed time $n\Delta t$, Riemannian (solid) vs KL (dashed). Rows: λ; columns: convergent stepsize Δt. The two schemes coincide at the fine stepsizes (left) but separate at the coarse stepsizes (right), where the Riemannian update reaches machine precision while KL plateaus higher.

$\Delta t\, C_n > 2$ regardless of how the covariance is advanced. This is why the two schemes' SPD and stability boundaries coincide on the quartic target and why the absolute stepsize ceiling is set by the mean, not the covariance.

- **At a matched stepsize the Riemannian scheme usually converges faster.** On both globally smooth targets the geodesic update reaches a smaller terminal energy gap than KL at every convergent stepsize, with the margin growing from a small factor at fine Δt to 10^4–10^8 at the coarsest convergent stepsizes (Sec. 6, Table 3); KL leads only on the non-smooth quartic and only modestly. So when one operates at a stepsize chosen as large as stability allows—the practical regime—the Riemannian update is typically the faster choice.

We are careful not to overclaim. At well-conditioned settings the absolute stable stepsize of the two schemes is comparable (indeed at $\lambda = 1$ the Riemannian scheme tolerates a slightly larger stable Δt than KL), and on the non-smooth quartic target both are restricted to small stepsizes, where KL is the faster of the two. The conclusions are specifically about the *gap between the KL proof bound and KL practice* (large and systematic) and the *matched-stepsize convergence-speed advantage of the Riemannian update on smooth targets*, not a blanket statement that one scheme dominates the other in every respect.

Limitations. This is numerical evidence on $d \leq 2$ deterministic targets, not a theorem. The stepsize grid is coarse (factors of 2–2.5), so the reported empirical maxima are accurate only to within

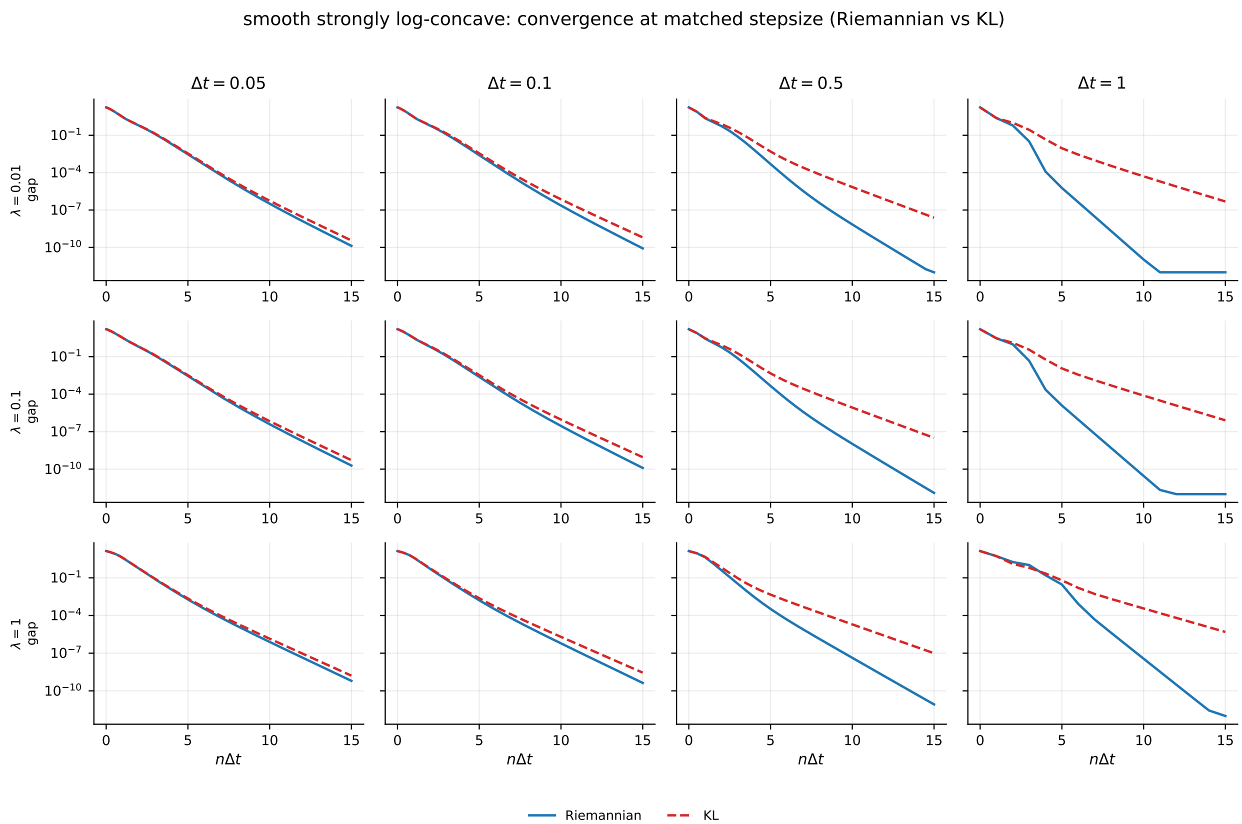

Figure 11: Smooth strongly log-concave posterior: energy gap vs elapsed time $n\Delta t$ (same layout and legend as Fig. 10). The same regime split holds: matched-stepsize agreement at fine Δt, and a growing Riemannian advantage as Δt coarsens.

one grid step; the qualitative four-to-six order gap is far larger than this resolution. Establishing that the KL scheme admits Riemannian-sized stepsizes in general requires a revised proof; the present data indicate where to look (the $\lambda_{\max}^3/\lambda_{\min}^3$ factor) and that the shared mean step, not the covariance update, is the true stepsize bottleneck.

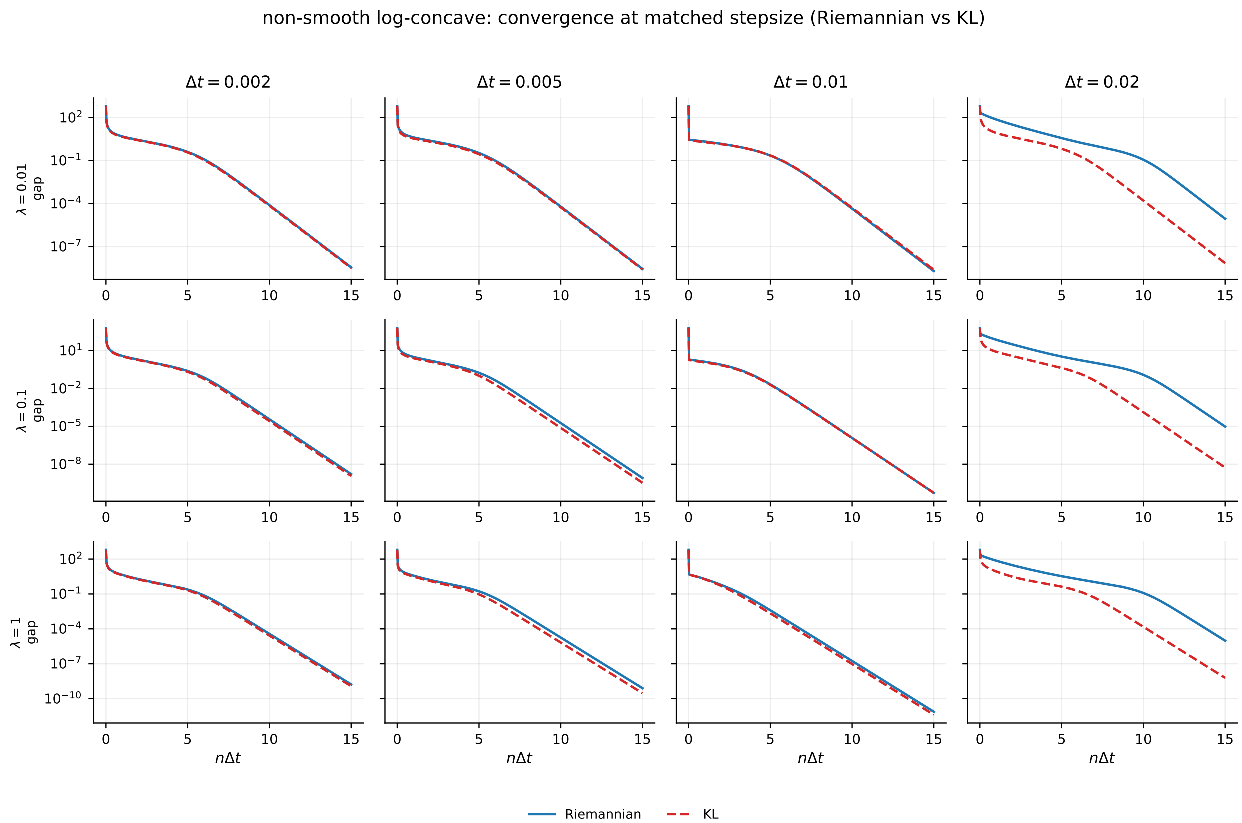

Figure 12: Non-smooth (quartic) log-concave posterior: energy gap vs elapsed time $n\Delta t$ (same layout). The convergent stepsizes are an order of magnitude smaller than for the smooth targets; the schemes agree closely at the finer stepsizes, but here *KL* holds the advantage—most clearly at the coarsest convergent $\Delta t = 0.02$, where the Riemannian exponential map sits near its stability edge (terminal gap $\sim 10^{-5}$) while KL has converged ($\sim 10^{-9}$). This is the one target on which KL, not Riemannian, is the faster scheme.

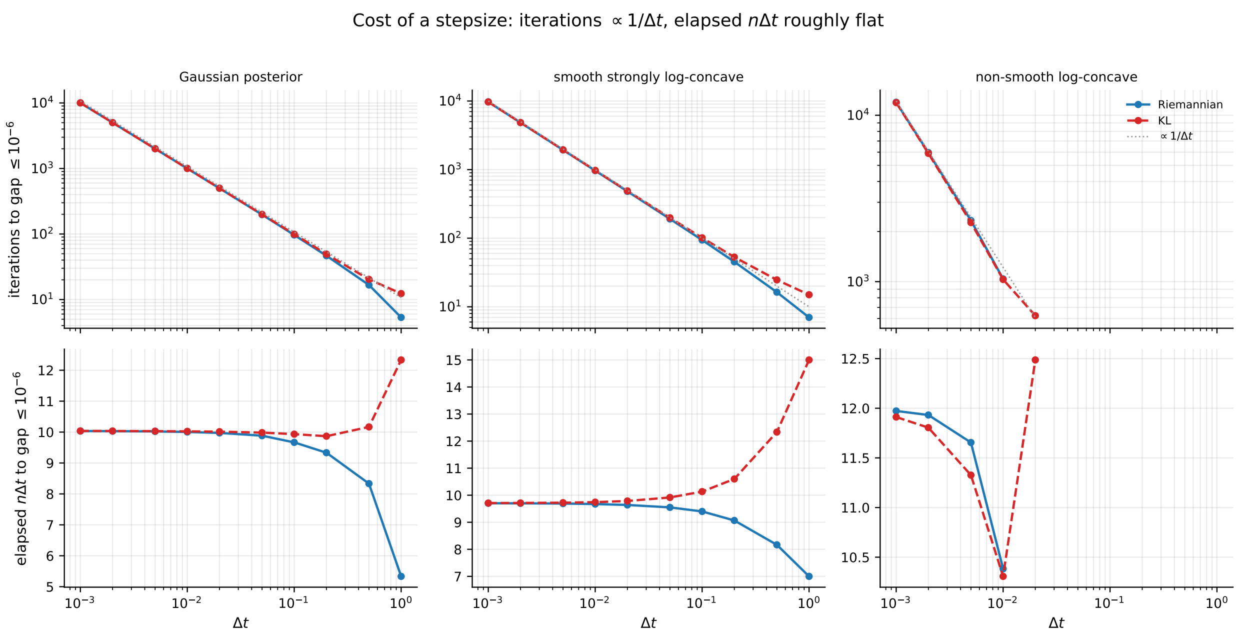

Figure 13: Cost to reach gap $\leq 10^{-6}$ for stable runs, averaged over λ, vs Δt. Top: iterations (log–log), with the $\propto 1/\Delta t$ reference slope; bottom: elapsed time $n\Delta t$ (semi-log x), nearly flat in Δt. Riemannian (blue) and KL (red) coincide at fine Δt and separate only at the coarsest convergent stepsizes; the faster scheme is target-dependent (Riemannian on the smooth targets, KL on the non-smooth one).